Deep CNN-Based Mango Insect Classification

Muntaka Rahat, Mehedi Hasan, Md. Mahmodul Hasan, Md. Tarequl Islam, Md. Syfur Rahman, A. K. M. Kamrul Islam, and Mohammad Motiur Rahman

1 Introduction

Mango insects are nowadays a big threat to the cultivation of mango. Attack of mango insects can have a severe outcome on the quality and quantity of mango production. A study on the impact of fruit fly on mango production indicated that production decreases from 24% to a higher loss of 60% [1]. If the insects can be classified adequately, then the cultivation of mango will give a significant improvement in

M. Rahat · M. Hasan · Md. M. Hasan · A. K. M. K. Islam · M. M. Rahman (✉)
Department of Computer Science and Engineering, Mawlana Bhashani Science and Technology University, Santosh, Tangail 1902, Bangladesh
e-mail: mm73rahman@gmail.com

M. Rahat
e-mail: muntakace15036@gmail.com

M. Hasan
e-mail: mehedicse15030@gmail.com

Md. M. Hasan
e-mail: mahmodul.mbstu@gmail.com

A. K. M. K. Islam
e-mail: kamrul3000@gmail.com

Md. T. Islam
Department of Information and Communication Technology, ICT Division, Dhaka, Bangladesh
e-mail: tareq07@gmail.com

Md. S. Rahman
Education Management Information System (EMIS) Cell, Ministry of Education, Dhaka, Bangladesh
e-mail: syfur022@gmail.com

A. K. M. K. Islam
Department of Computer Science, Georgia State University, Atlanta, GA 30302-5060, USA

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production. Almost a dozen insects have been found damaging the mangoes, which cause severe economic losses. The fruit fly, mango hopper, mango stone weevil, mealybug, stem borer, leaf weevil, red ant, and others are termed as the significant mango insects [2]. Distinguishing the proper category of insects is essential to take necessary actions. The diversity of these insects makes it rigid to classify a particular category correctly. However, a computer-aided solution can capture the diversity of insects to find a solution. This research proposes a scheme to classify mango insects.

In recent years, tremendous research has been conducted on insect classification based on advanced techniques such as machine learning or deep learning. However, there exist not many thorough studies on mango insect classification due to the unavailability of the dataset. Machine learning-based classification techniques require sophisticated datasets to perform analysis. However, most classification methods for image data use deep learning algorithms rather than traditional machine learning methods. The advances in computation resources make it feasible to use deep learning techniques instead of others. In this study, we also proposed a deep learning method to detect mango insects. This research experimented with different architectures of convolutional neural networks (CNN). Besides, we studied an aggregated technique known as an ensemble using three different fine-tuned CNN models. This study included the ensemble classification method to attain considerably better performance than individual models. We have also studied other architecture of the CNN model, while for the ensemble method, this chapter adopted VGG19, MobileNet, and Xception architecture. The reason for choosing these three architectures is to include different variants of convolution operation techniques applied in CNN models. This variation helps the ensemble method to achieve better performance. Here, the VGG19 uses general convolution operation with no optimization to reduce computational complexity. In contrast, both MobileNet and Xception use depthwise separable convolution, which is much more computationally efficient. However, the technique of depthwise separable convolution is a bit different in MobileNet and Xception.

The newly created dataset of this research contains three different types of insects *Dorsophila melanogaster* (Fruit Fly), *Idiocopus clypealis* (Mango Hopper), and *Sternochetus mangiferae* (Mango Stone Weevil). These species of insects are mostly affecting the mango production in the Asian region [3]. This research collects data from different open sources. Our newly created dataset is one of the main contributions of this research. The dataset contains a different train, test, and validation set with labels.

The contributions of this chapter are as follows:

- Ensemble-based deep learning methods for mango insect classification with improved performances. The ensemble includes fine-tuned VGG19, Xception, and MobileNet.
- The creation of a dataset for mango insect classification consists of the three most dangerous insects- stone weevil, fruit fly, and hopper.
- An intuitive study of deep learning methods for mango insect classification with limited data.

– A study of ensemble methods for performance improvement on insect classification.

The rest of the chapter organized as Sect. 2 presents previous studies related to this research. The ensemble method and pre-trained models fine-tuning are discussed in Sect. 3. Section 4 describes the performance of our proposed method. The last section of this chapter concludes the research and indicates future scopes in mango insect classification.

2 Related Works

Automatic insect classification and detection is an ongoing research topic for the last few years. In order to enhance the classification rate of insects, researchers have studied many techniques using machine learning, pattern recognition, and computer vision. Several machine learning and deep learning techniques have been studied for insect classification and detection. Computer vision and pattern recognition made substantial advances in previous years. Large Scale Visual Recognition Challenge (ILSVRC) based on ImageNet public dataset has been customized as a benchmark for different recognition related problems in computer vision, together with object classification and object identification. Despite the advances in insect detection, there are not many studies on mango insects. This section describes several studies related to dataset collection and insect classification and detection methods in recent years.

Authors of [4] designed fruit fly larvae detection on mango using a hyperspectral imaging technique. They used traditional techniques of image processing to detect and localize fruit fly larvae in mangoes. They claimed 87.7% accuracy of this technique. Xia et al. [5] utilized fine-tuned VGG19 to extract features from insects with a regional proposal network to classify and localize insects. Their proposed method achieved an mAP value of 89.22%. A fusion method of saliency image preprocessing and convolutional neural networks is studied in [6]. They used three different types of preprocessing to extract saliency images and used those images as input for CNN to classify insects with an accuracy of 61.93% in IP102 dataset for insect detection. Researchers of [7] designed a study on insect detection using a saliency map of the image with SIFT-HMAX and LCP feature extraction technique and SVM for classification. The proposed method of [7] can classify within 146.3 s with an accuracy of 85.5%.

Khalifa et al. [8] proposed a transfer learning method for insect classification on IP102 dataset. They used pretrain AlexNet model with an augmentation technique to classify insects into different classes. Their proposed method achieved an accuracy of 89.33% on IP102 test dataset. This method does not provide any comparison with recent advanced deep learning methods. The authors of [9] designed a convolutional network-based feature extraction method to identify the taxonomy of insects. They have used VGG16 network in their study for the feature extraction process. Fuentes et al. [10] proposed a real-time tomato insect detection scheme using deep

learning techniques. They combined three different object detection and recognition techniques in their "deep learning meta-architectures." Their work also extended to detect different diseases caused by insects. Besides, the proposed method worked on the localization of disease and insects.

Insect classification using a non-image-based technique is also studied by researchers using machine learning and traditional classification techniques. Phung et al. proposed an insect classification method using the acoustic feature of insects using the bagged tree and kNN classifier [11]. Their proposed classification method on eleven different types of insects achieved an accuracy of 92.3% using a bagged tree while classification accuracy using kNN was 88.5%. Saranwong et al. studied a near-infrared imaging technique for fruit fly larvae [12]. They performed this technique on intact mangoes. The drawback of this work is they worked with one particular type of insects.

3 Methodology

The methodology section describes the working procedure of our mango insects classification method. This research prepares a specific dataset of mango insects alongside a classification scheme. The first subsection illustrates the overall procedure of dataset preparation. The dataset preparation consists of several steps like data collection, cleaning, labeling, and decomposing, to make the dataset consistent. The next step of the methodology is model preparation. The model preparation step includes data augmentation, model training, and the ensemble of models. Figure 1 presents an overall view of the proposed methodology.

3.1 Dataset Preparation

Dataset preparation is the preliminary task of organizing data for training and evaluating machine learning or deep learning models. Appropriate data preparation aids model to find optimal parameters. Dataset preparation is a complex process to be usable for training deep learning models. The procedure includes defining the problem, and here it is mango insect classification from insect images. The next step towards this procedure is the data collection. The procedure includes sources of data, collection procedure, type of data to collect, and others [13]. This research uses Google image search to collect data from open-source image collections. The unqualified images are removed to maintain consistency in the dataset. The collected data needs to be in a specific format. This research uses a conversion mechanism for formatting images into JPEG format. The images are also resized into a standard dimension of 224×224.

The clean and consistent data are then distributed into train, validation, and test set. The data distribution is a significant part of dataset preparation. The dataset is first

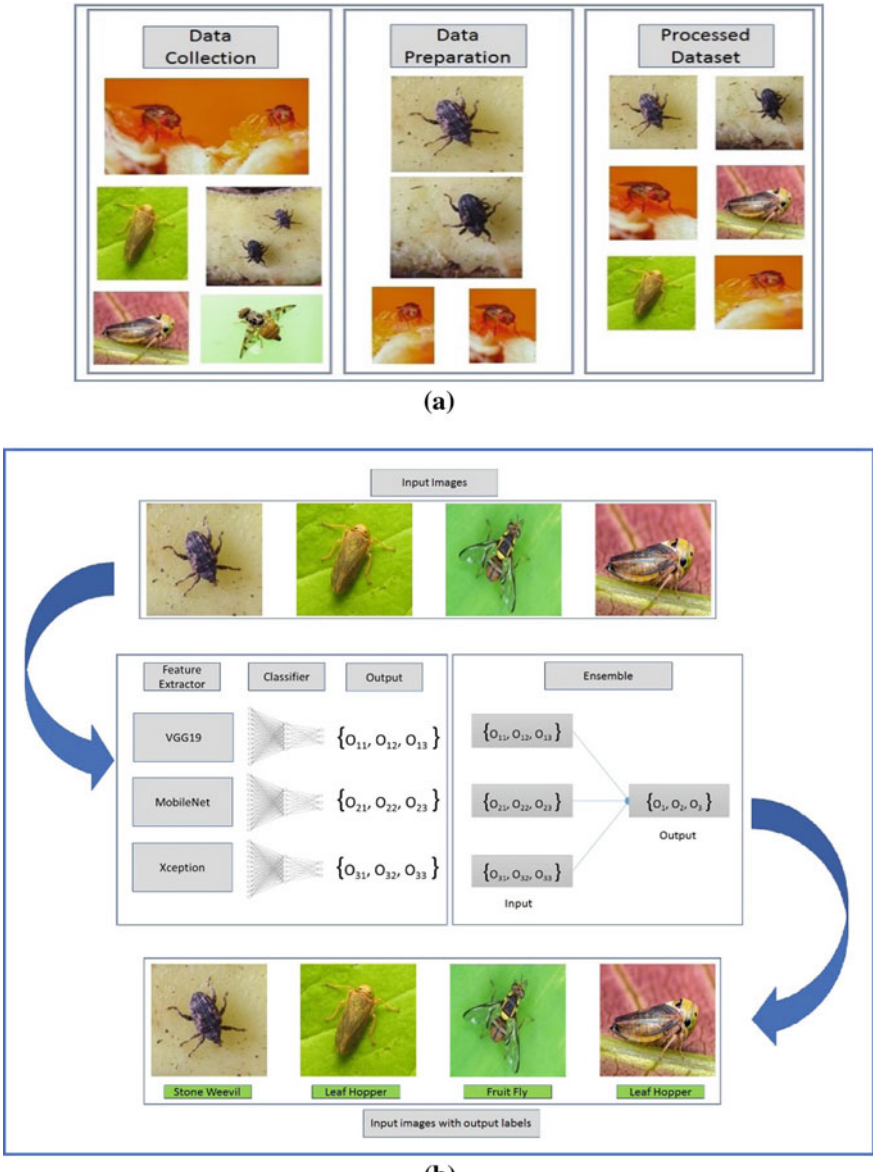


Fig. 1 **a** Dataset preparation process. **b** Block diagram of the proposed methodology

Table 1 The distribution of data among three classes (fruit fly, hopper, and stone weevil)

Species	Number of samples
Fruit fly	1215
Hopper	892
Stone weevil	402

divided into two sets- train and test where 85% of the clean data belongs to train set using random selection technique. Then the two datasets are checked for duplicates. The duplicate checking ensures the consistency of data distribution among train and test sets. After that, five percent of the total data is chosen randomly from the train set and make them a validation set. Again, a repetition check is performed on a new train and validation set and taken proper action to remove repetition.

Table 1 shows the distribution of data among different classes. This table presents the overall data distribution. The train set of the whole dataset contains 1215 fruit fly images, 892 hopper images, and 402 stone weevil images. There are 60, 45, and 20 images for fruit fly, hopper, and stone weevil, respectively in the validation set and test set consist of the remaining of the data with 180 images of the fruit fly, 135 hopper images, and 60 images of stone weevil. Table 2 presents the data distribution among train, validation and test set. Figure 2 shows a few instances of different insects belong to this dataset.

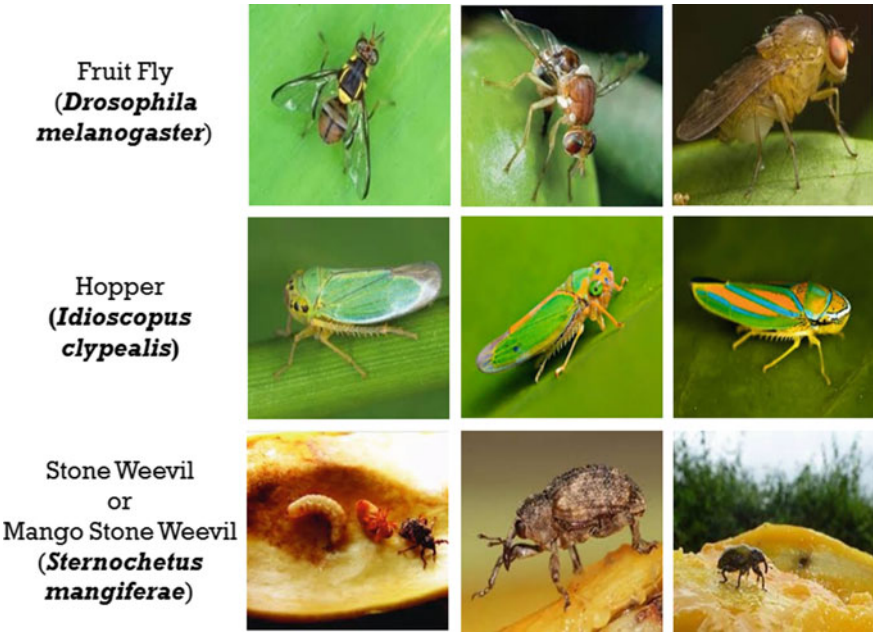


Fig. 2 A few instances of fruit fly, hopper, and mango stone weevil

3.2 Ensemble-Based Classification

Classification is a term popularly used in deep learning, and the machine learning domain refers to the system to categorize data into classes. Herein, we studied mango insect classification using an ensemble technique. The ensemble is a fancy term to represent the aggregation of several methods that work together to find an optimal solution. This study uses an ensemble of three different pre-trained CNN model architecture to classify mango insects. A model that is already trained on a dataset and that weights of that model can be used to solve another problem. This already trained model is known as a pre-trained model. This work uses pre-trained models to get benefits from extensive training of models on large datasets like ImageNet. The three models are Xception, MobileNet, and VGG19. Pre-trained models are not ready to go for every classification; these models require some modification and retraining to be useful for particular purpose. The procedure of retraining a pre-trained model for any particular task is referred to as fine-tuning the model. We have fine-tuned these models as required for insect classification by tuning parameters and hyper-parameters of these models. The fine-tuning of these models are discussed in the next part of this section. In order to compensate with a limited dataset, we have utilized data augmentation techniques to increase the dataset size.

Data Augmentation: Data augmentation is a process to create different variants of the same data to increase diversity in the dataset. Data augmentation is a handy technique to work with limited data. The augmentation of data requires to maintain some constraints to be useful for classification. This work used augmentation techniques to increase dataset size artificially. Most of the mango insects are small in size, thus zooming images to a certain extend is convenient. This study used random zooming of up to 0.2. This random zooming increase diversity while training and reduce model overfitting. Besides, the training procedure also used shearing with randomly chosen value within a range of 0 to 0.1. Flipping is another augmentation technique; we have applied during training.

Moreover, complex image augmentation, like random rotation with zooming, also applied. The pre-trained models require scaled images with pixel values within a range of 0.0 to 1.0. Each image of the dataset is rescaled by dividing the pixel values with the highest intensity value of 255. Fine-tuning of pre-trained models for insect classification is discussed in the next few subsections.

Fine-Tuned VGG19 Model: Training a nineteen weighted layers VGG19 architecture from scratch requires a large amount of data. A trained model with limited

Table 2 Train-validation-test dataset distribution

Dataset	Fruit fly	Hopper	Stone weevil
Train	975	712	322
Validation	60	45	20
Test	180	135	60

data fails to give a good performance. Instead of using a scratch model for limited data, retraining a trained model with vast amounts of data helps to attain considerable performance [14]. This retraining of existing pre-trained is beneficial as these models capable of learning low-level features of any image. Due to a lack of extensive data, we have adopted a pre-trained VGG19 model for classification. The pre-trained VGG19 was previously trained on the ImageNet dataset contains nineteen layers. The pre-trained model can classify images into 1000 classes. However, this research requires images to classify into three categories. To use a pre-trained VGG19 model, we need to fine-tune as our requirement for three-class classification.

The fine-tuning procedure includes several changes in the classifier part of the model. The output layer of the fine-tuned model contains three neurons instead of 1000 neurons of the pre-trained model. This is a three-class (fruit fly, hopper, and stone weevil) classification problem; thus, the fine-tuned model contains three neurons instead of a thousand neurons of the pre-trained model. The model with modified output layer performance is surpassed by a lower number of hidden neurons in hidden layers. After experimenting with several parameters, this research used 512 and 256 neurons on hidden layers with dropout. Besides, dropout facilitates the random presence and absence of each neuron with a certain probability. The model uses the softmax activation function at the output layer and the ReLU activation function at hidden layers. The batch size of the model for training was 64 images of size 224 $\times$ 224. The weight of the convolution part of the pre-trained model is not trained, while the classifier part is randomly initialized and optimized using the mango insect dataset. Figure 3 shows the architecture of the fine-tuned model.

Fine-Tuned MobileNet Model: The MobileNet model uses a different technique for convolution rather than the convolution technique applied in VGG19. The depthwise

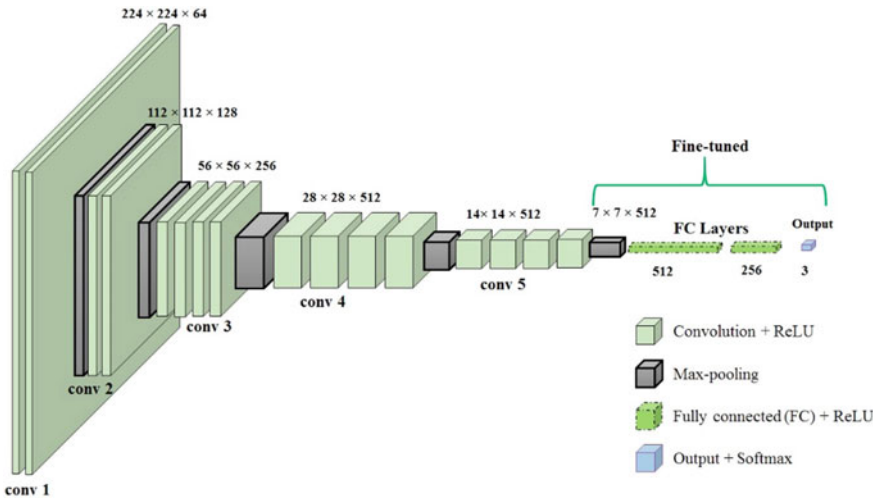


Fig. 3 The architecture of fine-tuned VGG19

Separable Convolution (DSC) technique of the MobileNet model makes it efficient to train with less computing resources without compromising the performance [15]. To utilize the MobileNet pre-trained model for mango insect classification, we need to fine-tune the model.

The fine-tuning process of MobileNet architecture eliminates the classifier head of the architecture and redesigns the classifier part of the model as required to classify mango insects. The experimented dataset contains three different classes thus, the fine-tuned model has three units at the output layer. These three output units or neurons are responsible for classifying an input image to one of the three classes. The average pooling layer of MobileNet architecture is replaced with a Global Average Pooling (GAP) to decrease the fine-tuned model parameters. This layer is followed by two fully-connected layers with 512 neurons on the first hidden layer and 256 neurons on the second. Batch-normalization is applied to speed up the training process. The experiments use dropout to eradicate the effect of over-fitting each hidden layer followed by a dropout layer in this regard. The network uses the ReLU activation function for hidden layers. Figure 4 shows the fine-tuned architecture of the MobileNet model with depthwise convolution visualization.

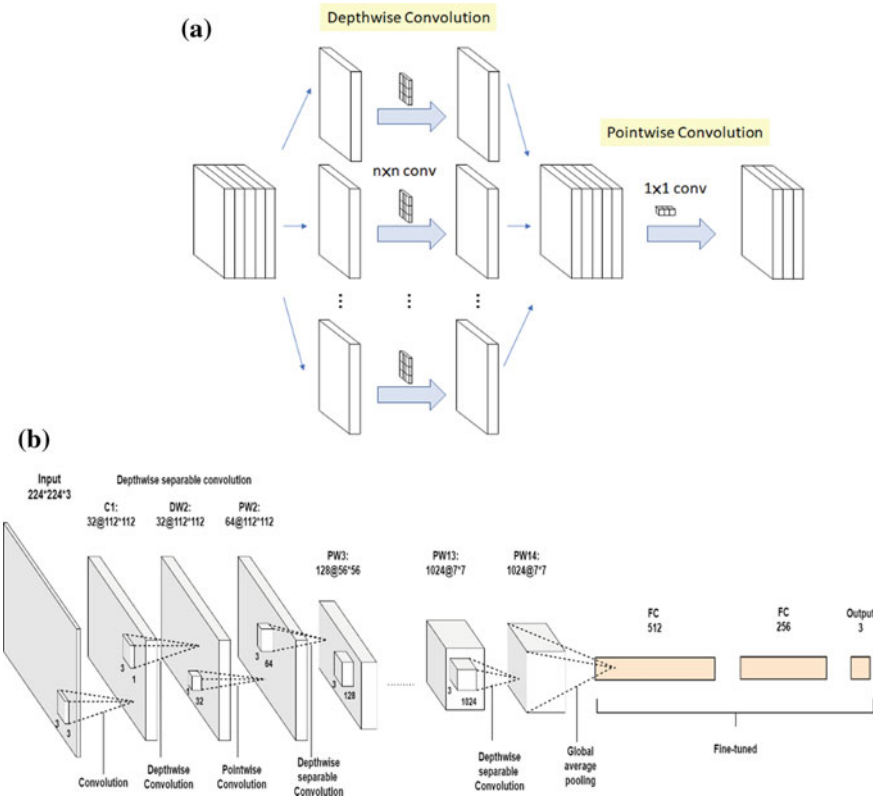


Fig. 4 **a** Depthwise separable convolution of MobileNet and **b** Fine-tuned MobileNet architecture

The output layer uses the softmax activation function. This fine-tuned model uses Adam optimizer to update weights. The early stopping technique is also utilized to stop the training process when validation accuracy remained constant for a few epochs. This technique helps to train the model without overfitting. The learning rate of training is set to 1e-4 at the initial stage of model compilation.

Fine-Tuned Xception Model: The architecture of Xception uses a different form of DSC. The procedure of DSC uses the reverse operation strategy applied in MobileNet DSC. The DSC of Xception performs pointwise convolution, followed by depthwise convolution. This procedure of DSC helps the Xception model to train faster than others with improved performance [16]. Figure 5 illustrates the DSC of Xception with the fine-tuned architecture of Xception.

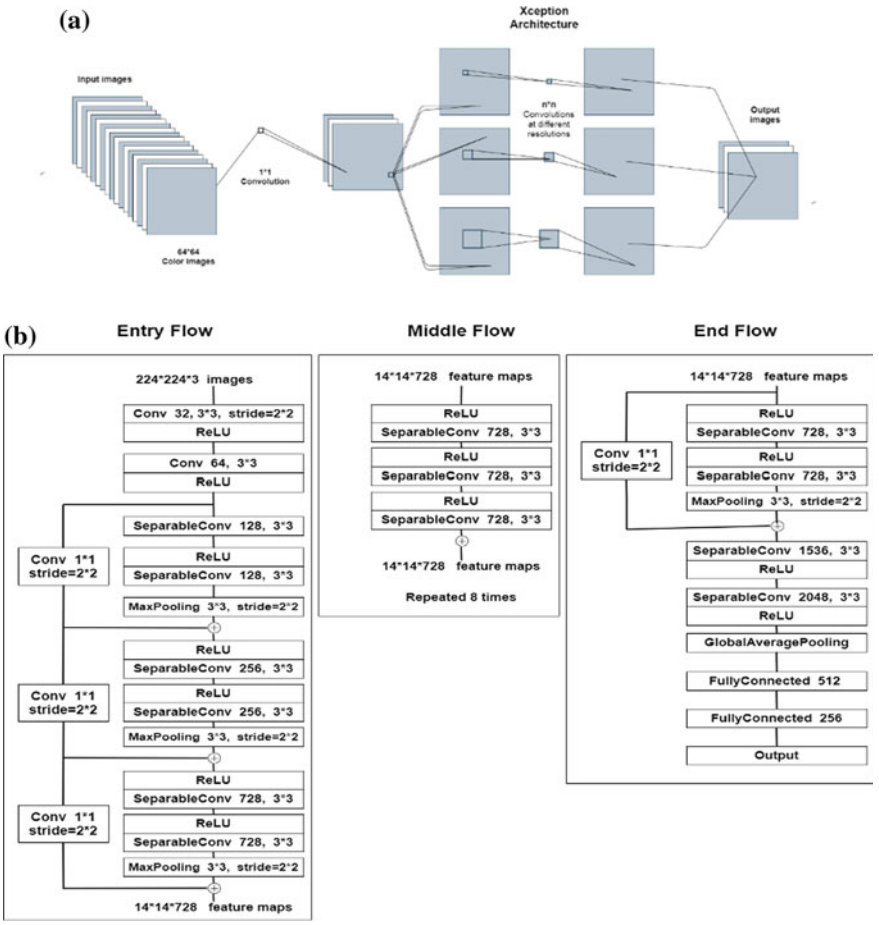


Fig. 5 **a** Depthwise separable convolution of Xception and **b** fine-tuned Xception architecture

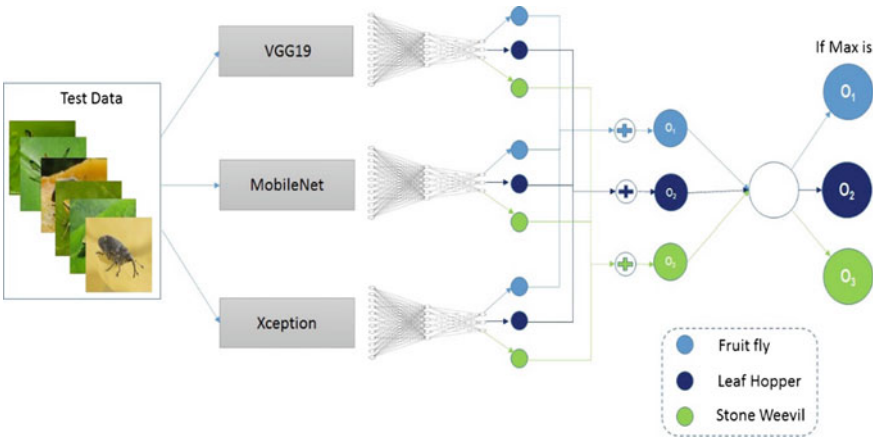


Fig. 6 Ensemble the procedure of the proposed method

The fine-tuning process of the Xception model is similar to two other models with few differences. The output layer of the model has the same number of neurons as the previous two models. The difference between the previous fine-tuning process with this tuned process is that this does not apply GAP as pre-trained already has GAP. We have used two hidden layers with a similar number of neurons as previous models. Batch-normalization is applied after the GAP layer. The hidden layer uses ReLU activation, while the output layer uses softmax activation. Dropout is applied to reduce overfitting. The fine-tuned model also uses Adam optimizer and accuracy as metrics. Early stopping is also utilized as previous models.

The ensemble of Models: The ensemble method is an excellent way of combining results from different methods to achieve improved performance in classification. This proposed method uses the result of the previously mentioned three fine-tuned models. This ensemble method uses the classification output of each model and sums them for each class. The class with the highest sum is the output of the ensemble method. The implemented algorithm uses five digits after the decimal point to minimize tie. Figure 6 shows a graphical illustration of the ensemble method using three fine-tuned model. The ensemble method helps attain better results than a single classifier. This chapter also experimented with other types of ensemble mechanisms such as voting where the final result is the class with the maximum number of votes. However, the summing of probabilities gave a significant performance. Algorithm 1 illustrates the algorithmic classification procedure of mango insects using the ensemble technique.

Algorithm 1 Ensemble technique for classification of mango insects.

Input: Images of mango insects with actual labels as $\{(x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m)\}$, here x_i represents a test image and y_i is the corresponding actual label of the image for $i = 1, 2, 3, \ldots, m$. (m is the total number of test images).

Output: Predicted output (y_pred_i) of images using ensemble classification for $i = 1, 2, 3, \ldots, m$.

begin
Design N models for the intended task $j = 1, 2, 3, \ldots, N$.
for $j = 1$ to N **do**
Train the jth model on the training dataset and find the optimum parameters by utilizing the performance on the validation set.
Store the optimal model as M_j for testing. (M is the collection of N models)

for $i = 1, 2, 3, \ldots, m$ test images **do**
initialization: $max_{prob} = 0$, prob_sum = {0}
for models M: $j = 1$ to N **do**
Give test image x_i as input to the model M_j
for k in 1, 2, …, number of classes (n_c) **do**
$prob_sum_k$ += value of kth output neuron of M_j
if $prob_sum_k > max_{prob}$ **then**
$max_{prob} = prob_sum_k$
$y_pred_i = k$

Evaluate the performance of ensemble classification using y_i and y_pred_i for i in 1 to m.
end

The algorithm illustrates that the ensemble procedure starts with an image input. Performance evaluation on test images with labels is symbolized as (x_i, y_i) where x_i is the image and y_i is the actual label of the image. Each model outputs the probability value of that image for different classes. The class with the highest probability value is the predicted class of that image. The ensemble technique utilizes the probability value of every class from each model and aggregates the probability of every class in an array. The aggregated probability is stored in *prob_sum* array for the number of total classes (n_c). ith index of *prob_sum* array represents the probability sum of ith class among n_c classes. This aggregation procedure of the ensemble technique overcomes the downfall of using a single model for classification. The maximum value is also calculated while computing the cumulative value of the probability to find the output class.

4 Results and Discussion

The results of mango insect classification are discussed in this section. Besides, this section also presents the result of data collection and preparation of the mango insect dataset. Subsequent subsections present and discuss the results of each fine-tuned model. Section 4.2 demonstrates a detailed performance analysis of the ensemble method.

4.1 Performance of Fine-Tuned Models

VGG19 Performance: Fig. 7 illustrates the performance of the fine-tuned VGG19 model on test data. We can calculate the accuracy of the model from the confusion matrix of the VGG19 model on test data. The accuracy of multiclass classification is calculated using Eq. (1). It is more informative to calculate the precision and recall score of the model on the test dataset. Equations (2) and (3) present the calculating formula for precision and recall. The precision score of the fine-tuned VGG19 model on test data is 89% for fruit fly, 99% for hopper, and 82% stone weevil. The recall score is also known as sensitivity. Recall or sensitivity score of fine-tuned VGG19 is 93%, 92%, and 85% for fruit fly, hopper, and stone weevil accordingly. The overall accuracy of the model on the test dataset is 91%. Table 3 presents the performance metrics of the fine-tuned VGG19 model.

$$\text{Accuracy} = \frac{\text{correctly classified samples } (n_c)}{\text{total number of samples } (n)} \tag{1}$$

$$\text{Precision} = \frac{\text{correctly classified positive samples } \left(n_{tp}\right)}{\text{total number of predicted positive samples } \left(n_{tp+}n_{fp}\right)} \tag{2}$$

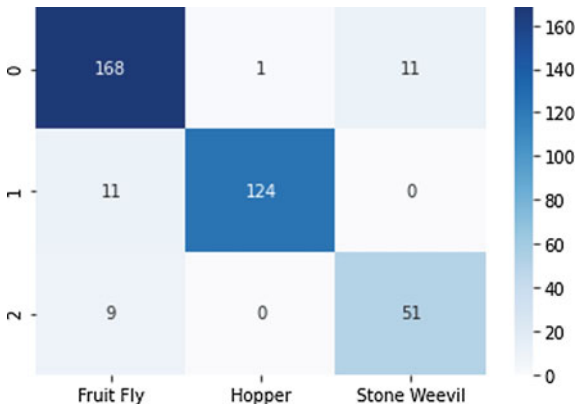


Fig. 7 Confusion matrix of fine-tuned VGG19 model

Table 3 Performance metrics of fine-tuned VGG19 model on mango insect classification test data

Performance metric	Fruit fly	Hopper	Stone weevil	Macro average
Accuracy	0.91			
Precision	0.89	0.99	0.82	0.90
Recall	0.93	0.92	0.85	0.90
F1-score	0.91	0.95	0.84	0.90

$$\text{Recall} = \frac{\text{correctly classified positive samples } (n_{tp})}{\text{total number of actual positive samples } (n_{tp} + n_{fn})} \tag{3}$$

MobileNet Performance: The performance of MobileNet is also calculated on test data. Figure 8 shows the confusion matrix for MobileNet on test data. MobileNet attains 97% precision on fruit fly, 94% on the hopper, and 96% on the stone weevil. The recall score of MobileNet on test data is interesting as it gives 99% on fruit fly and hopper classification. However, MobileNet attains a comparatively low recall score on stone weevil classification with a sensitivity value of 80%. Recall and sensitivity are the same measures with different names. This indicates that the architecture of MobileNet fails to identify stone weevil correctly. Table 4 presents performance metrics.

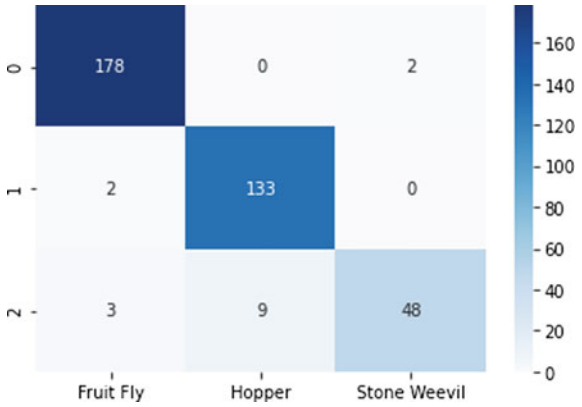


Fig. 8 MobileNet performance on test dataset

Table 4 Performance on test data of fine-tuned MobileNet model

Performance metric	Fruit Fly	Hopper	Stone weevil	Macro average
Accuracy	0.96			
Precision	0.97	0.94	0.96	0.96
Recall	0.99	0.99	0.80	0.92
F1-score	0.98	0.96	0.87	0.94

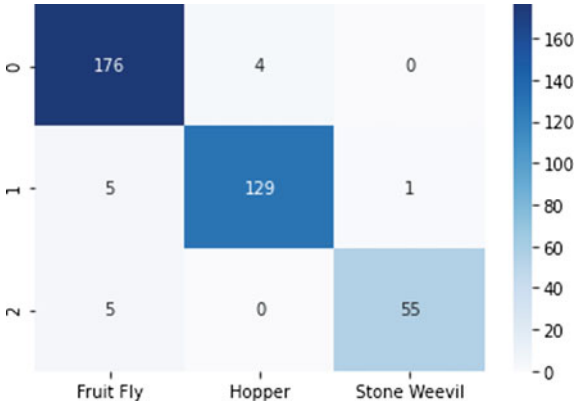


Fig. 9 Confusion matrix of Xception model on test dataset

Table 5 Performance metrics of fine-tuned Xception model on mango insect classification test data

Performance metric	Fruit fly	Hopper	Stone weevil	Macro average
Accuracy	0.96			
Precision	0.95	0.97	0.98	0.97
Recall	0.98	0.96	0.92	0.95
F1-score	0.96	0.96	0.95	0.96

Xception Performance: Fine-tuned Xception gives a better performance than the fine-tuned VGG19 and MobileNet model. The performance accuracy of the Xception model on the test set is 96%. Figure 9 shows the confusion matrix of the Xception model's performance on the test dataset. Table 5 shows the detailed performance metrics for the fine-tuned Xception model. The sensitivity or recall of each class is 98%, 96%, and 92% respectively for fruit fly, hopper, and stone weevil. The precision score of the Xception model on the test dataset is 95% for fruit fly, 98% on stone weevil, and 96% for hopper images.

4.2 Ensemble Method Performance

The ensemble method aggregates the results of fine-tuned models and uses the aggregated result for classification. The aggregated results help the ensemble method to boost the performance of classification. This aggregation of results helps to overcome the shortcoming of the individual model. Figure 10 presents the confusion matrix of mango insect classification using an ensemble method. The ensemble method has attained the best precision score and sensitivity score on each model. The precision score of the ensemble method on the test set is as follows 96% for fruit fly, 99% on hopper images, and flawless on stone weevil images. Also, the recall score on the ensemble method is quite good on different images. The recall score of fruit

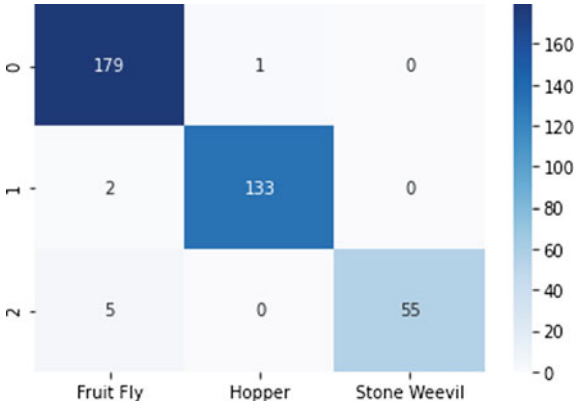


Fig. 10 Performance of ensemble method on test set (Confusion matrix)

Table 6 Performance metrics of ensemble method on test data

Performance metric	Fruit fly	Hopper	Stone weevil	Macro average
Accuracy	0.98			
Precision	0.96	0.99	1.00	0.98
Recall	0.99	0.99	0.92	0.97
F1-score	0.98	0.99	0.96	0.97

fly and hopper images is 99%, while 92% on stone weevil images. The accuracy of the ensemble method on test data is better than each model with an accuracy of 98%. Table 6 shows the performance metrics of ensemble method.

Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC): These are two significant metrics for evaluating classification performance [17]. Figure 11 represents the ROC curve with the AUC value of different classes for ensemble classification. The figure also represents the micro and macro average of the ROC curve with the AUC score. Hopper images achieved the best AUC score of 0.99, and stone weevil has the least AUC score. Both micro and macro average [18] AUC score of the ensemble method is 0.98. The interpretation of the AUC score indicates that the higher the classification model's value, the better the model is for classification. From this interpretation, it is evident that this proposed method performs well on mango insect classification.

Precision-Recall (PR) curve: The PR curve presents the precision and recall value of the classification model at different thresholds. The PR curve is more important for methods that work on an imbalanced dataset. The imbalance between the number of images in the largest class and smallest class in our dataset is not significantly high. PR curve gives significantly better intuition on the moderate class imbalance. Figure 12 shows the PR curve of our proposed method.

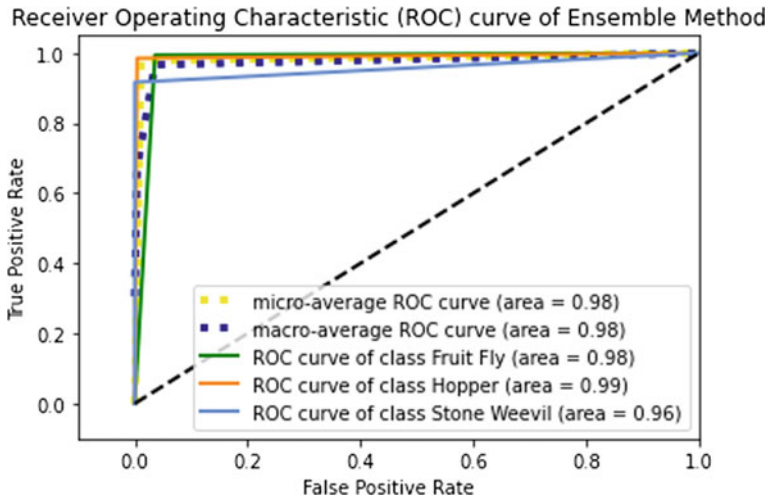


Fig. 11 ROC curve with AUC score of ensemble method

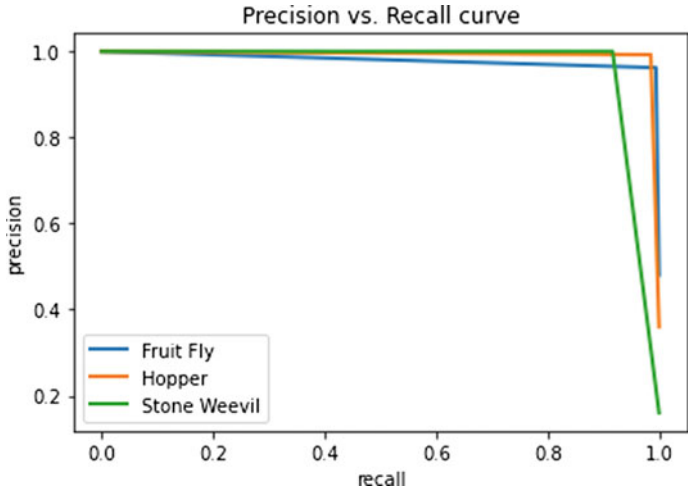


Fig. 12 Precision versus Recall curve of ensemble method

5 Conclusion

The adverse of insects in mango production cause substantial economic loss. The use of pesticides without knowing the category of pests can cause severe damage to the growth of mango. This chapter proposed a mango insect classification scheme using an ensemble of CNN models. Besides, this study developed a dataset of mango insects to aid automated mango insect classification. The dataset contains three thousand

clean images with labels distributed into train, validation, and a test set of three different kinds of mango insects named mango stone weevil, mango hopper, and fruit fly. This research also presented a study of ensemble techniques to classify mango insects. The ensemble technique used three well-performed fine-tuned CNN models to boost the performance of the ensemble technique with an accuracy of 98%. This work will be extended to increase the dataset size with several types of mango insects in the future. It will be interesting to use other techniques to detect insects in real-time.

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